Math 472: Homework 06

Due Wednesday, March 11

Problem 1 (Exercise 8.50)**.** Refer to example 8.8 in the textbook. In this example, p_1 and p_2 were used to denote the proportions of refrigerator of brands A and B, respectively, that failed during the guarantee periods.

(a) At the approximate 98% confidence level, what is the largest "believable value" for the difference in the proportions of failures for refrigerators of brands A and B?

(b) At the approximate 98% confidence level, what is the smallest "believable value" for the difference in the proportions of failures for refrigerators of brands A and B?

(c) If $p_1 - p_2$ actually equals 0.2251, which brand has the larger proportion of failures during the warranty period? How much larger?

(d) If $p_1 - p_2$ actually equals -0.1451, which brand has the larger proportion of failures during the warranty period larger? How much larger?

(e) As observed in Example 8.8, zero is a believable value of the difference. Would you conclude that there is evidence of a difference in the proportions of failures (within the warranty period) for the two brands of refrigerators? Why?

Problem 2 (Exercise 8.70)**.** Let Y be a binomial random variable with parameter p. Find the sample size necessary to estimate p within .05 with probability .95 in the following situations:

- If p is thought to be approximately .9
- If no information is known about p (use $p = .5$ in estimating the variance of $\widehat{p}$).

Problem 3 (Exercise 8.81)**.** The carapace lengths (in mm) of ten lobsters examined in a study of the infestation of the *Thenus orientalis* lobster by two types of barnacles, *Octolasmis tridens* and *Octolasmis lowei*, are as follows:

$$78 \quad 66 \quad 65 \quad 63 \quad 60 \quad 60 \quad 58 \quad 56 \quad 52 \quad 50.$$

Find a 95% confidence interval for the mean carapace length (in mm) of *Thenus orientalis* lobsters.

Problem 4 (Exercise 8.85)**.** Two new drugs were given to patients with hypertension. The first drug lowered the blood pressure of 16 patients an average of 11 points, with a standard deviation of 6 points. The second drug lowered the blood pressure of 20 other patients an average of 12 points, with a standard deviation of 8 points. Determine a 95% confidence interval for the difference in the mean reductions in blood pressure, assuming that the measurements are normally distributed with equal variances.

Problem 5 (Exercise 9.1)**.** Suppose that Y_1, Y_2, Y_3 are a random sample from an exponential distribution with density function

$$f(y) = \frac{1}{\theta} e^{-y/\theta} \mathbf{1}_{[y>0]}$$

Consider the following estimators

$$\hat{\theta}_1 = Y_1, \qquad \widehat{\theta}_2 = \frac{Y_1 + Y_2}{2}, \qquad \widehat{\theta}_3 = \frac{Y_1 + 2Y_2}{3}, \qquad \widehat{\theta}_4 = \min(Y_1, Y_2, Y_3), \qquad \widehat{\theta}_5 = \overline{Y}$$

In problem #8 of homework 4, you showed that $\widehat{\theta}_1, \widehat{\theta}_2, \widehat{\theta}_3$ and $\widehat{\theta}_5$ are unbiased estimators of θ.

(a) Find $\text{eff}(\widehat{\theta}_1, \widehat{\theta}_5)$.

(b) Find $\text{eff}(\widehat{\theta}_2, \widehat{\theta}_5)$.

(c) Find $\text{eff}(\widehat{\theta}_3, \widehat{\theta}_5)$.

Problem 6 (Exercise 9.6)**.** Suppose $Y_1, \ldots, Y_n$ is a random sample of size n from a Poisson distribution with mean λ. Consider $\widehat{\lambda}_1 = \frac{Y_1+Y_2}{2}$ and $\widehat{\lambda}_2 = \overline{Y}$. Derive the efficiency of $\widehat{\lambda}_1$ relative to $\widehat{\lambda}_2$.

Problem 7 (Exercise 9.8)**.** Let $Y_1, \ldots, Y_n$ denote a random sample from a probability density function $f(y)$, which has unknown parameter θ. If $\widehat{\theta}$ is an unbiased estimator of θ, then under very general conditions

$$\mathrm{Var}(\widehat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}$$

where

$$\mathcal{I}(\theta) = n\mathbb{E}\left[-\frac{\partial^2}{\partial\theta^2}\Big[\log(f(Y))\Big]\right].$$

(This is known as the Cramer-Rao inequality.) If $\mathrm{Var}(\widehat{\theta}) = \frac{1}{\mathcal{I}(\theta)}$, the estimator $\widehat{\theta}$ is said to be *efficient*.

Suppose that $f(y)$ is the normal density with mean μ and variance σ^2. Show that $\overline{Y}$ is an efficient estimator of μ.

Problem 8 (Exercise 9.17)**.** Suppose that $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ are independent random samples from populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively. Show that $\overline{X} - \overline{Y}$ is a consistent estimator of $\mu_1 - \mu_2$.

Problem 9 (Exercise 9.24)**.** Let $Y_1, \ldots, Y_n$ be independent standard normal random variables.

(a) What is the distribution of $\sum_{i=1}^n Y_i^2$?

(b) Let $W_n = \frac{1}{n}\sum_{i=1}^n Y_i^2$. Does W_n converge in probability to some constant? If so, what is the value of the constant?

Problem 10 (Exercise 9.39)**.** Let $X_1, \ldots, X_n$ denote a random sample from a Poisson distribution with parameter λ. Show that $\sum_{i=1}^n Y_i$ is sufficient for λ.